

Chapter 9: The number plane

6 lessons · one lesson for each exercise · each lesson is one hour

NSW syllabus outcome

Stage 4 Number and algebra: MA4-LIN-C-01 creates and displays number patterns and graphs linear relationships on the Cartesian plane.

CHAPTER

9

9.01 Grid references on maps

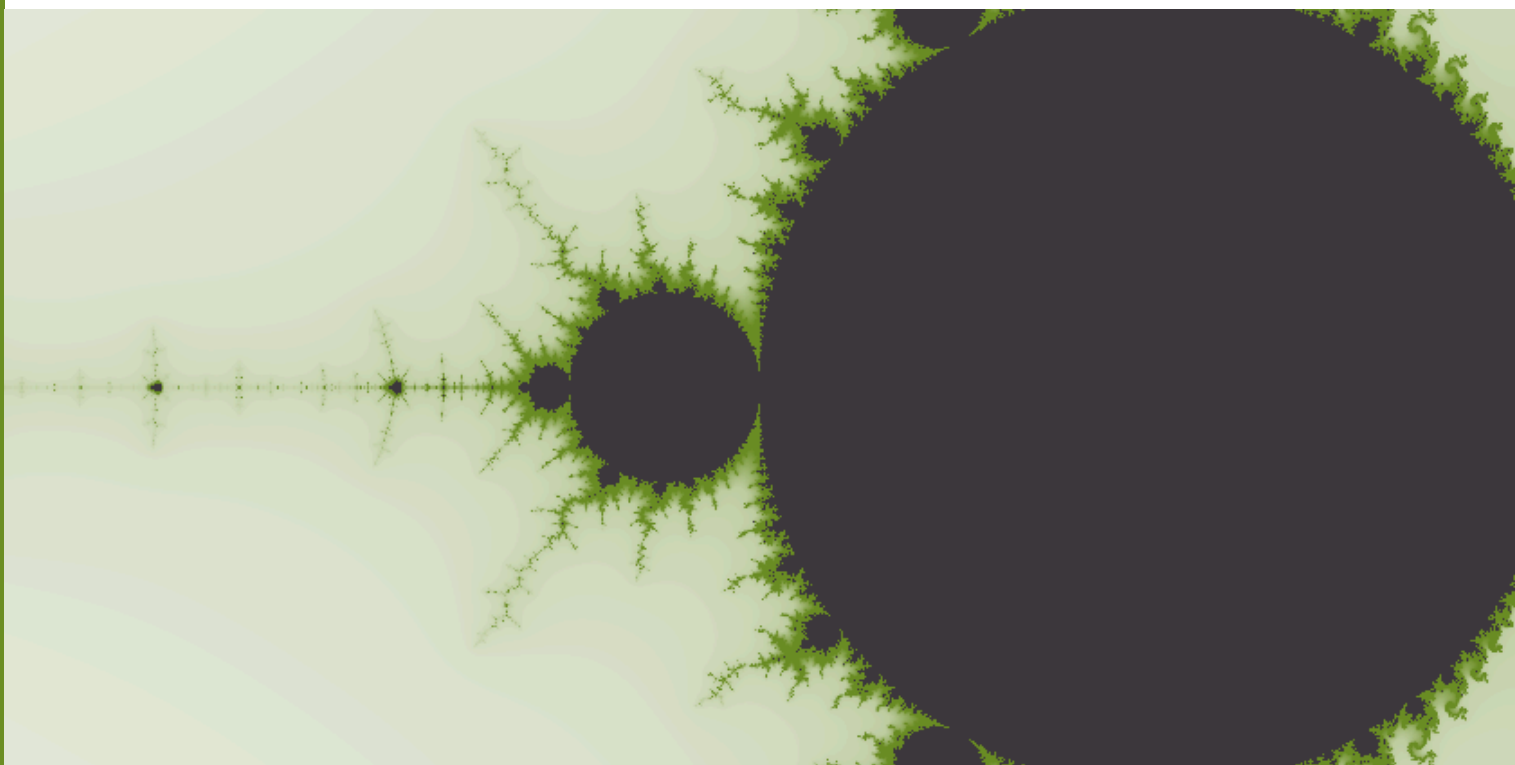
9.02 Points in the first quadrant

9.03 The number plane

9.04 Plotting tables of values

9.05 Tables of values from rules

9.06 Graphing linear equations



How to use this booklet

1 Each lesson is **one hour**. Start with the **learning intentions** and **success criteria**, write the **key terms** and use the **notes** box.

WE DO

Grey boxes. Your teacher works each example (parts a, b, c) on the board. **Copy every step** into the space.

3 Finished early? Try the **extension**. At the end, **show off your skill** on the tear-off exit ticket and hand it in.

2 Every lesson has three levels: **Easy**, then **Medium**, then **Challenging**. Work down the page: the questions get harder as you go.

YOU DO

Blue boxes. Now it is your turn. Work on your own. Quick drill answers go on the line; show working in the boxes.

4 Missed a lesson? Each lesson ends with a **summary** with a worked example for every level. Start there.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
9.01	Grid references on maps	3		☐
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9.04	Plotting tables of values	21		☐
9.05	Tables of values from rules	27		☐
9.06	Graphing linear equations	33		☐

In this chapter you will

- use grid references on a map
- plot and read points in the first quadrant
- plot and read points in all four quadrants
- plot points from tables and describe patterns
- complete tables of values from rules
- graph linear equations and test whether a point is on a line

About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
9.01

Grid references on maps

Date _____

Learning intentions

We are learning to:

- use grid references to find and describe locations on a map

Success criteria

I can:

- ☐ read a grid reference: letter (across) then number (up)
- ☐ state the feature at a grid reference
- ☐ write the grid reference of a feature

Key terms Write the definition as your teacher explains it.

Grid reference

Map

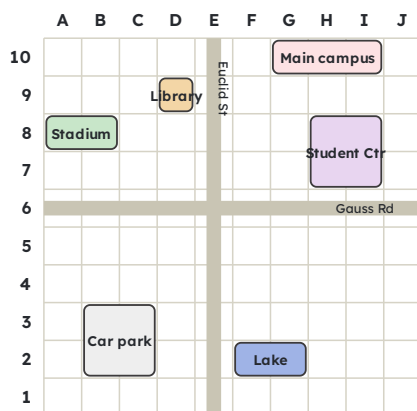
Notes

Level 1 · Getting started

Easy · 16 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Use the map. What is at each grid reference?



a What is at F2?

Your teacher works this out. Copy every step.

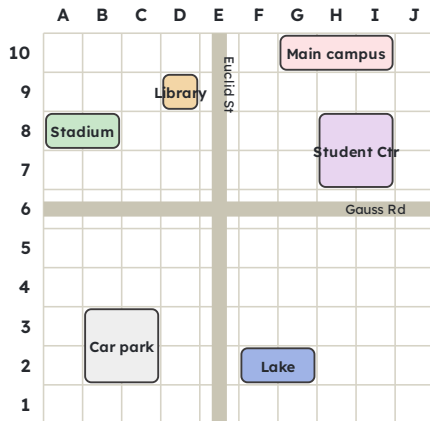
b What is at D9?

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Use the map. What is at:

4 QUESTIONS

REMEMBER A grid reference names a square.



1 F2

2 D9

3 A8

4 H7

YOU DO On your own: **Easy** Set A2. Quick drill: use the map in Set A. What is at:

6 QUESTIONS

1 B2

2 C3

3 G2

4 B8

5 I8

6 H10

YOU DO On your own: **Easy** Set B. Write a grid reference for:

6 QUESTIONS

1 the library

4 the car park

2 the lake

5 the student centre

3 the stadium

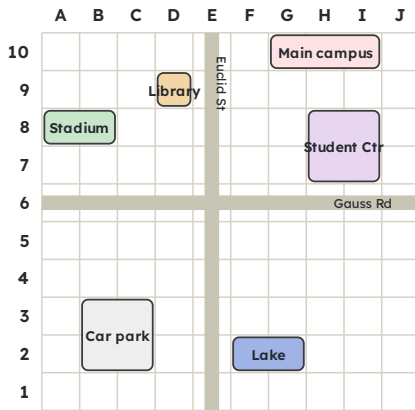
6 the main campus

Level 2 · Building up

Medium · 6 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Use the map. Write the grid references.



a Write the grid reference of the library.

Your teacher works this out. Copy every step.

b Which grid squares does the car park cover?

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Use the map.

6 QUESTIONS

REMEMBER Letter first (across), then number (up).

1 Name a grid square that Gauss Road passes through.

3 Where do Gauss Road and Euclid Street meet?

5 How many squares is the lake from the car park, across?

2 Name a grid square that Euclid Street passes through.

4 Which feature is furthest north (highest row)?

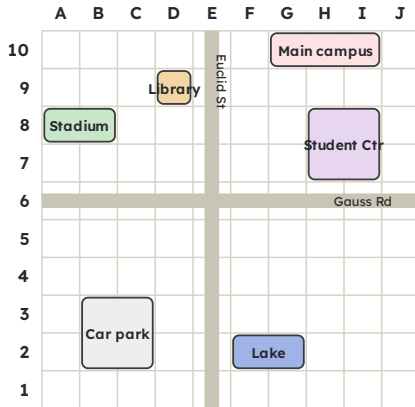
6 Which features are in column H?

Level 3 · Challenge

Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Use the map, then explain.



a Describe how to walk from the lake to the library along the roads.

Your teacher works this out. Copy every step.

b Why is the letter written before the number?

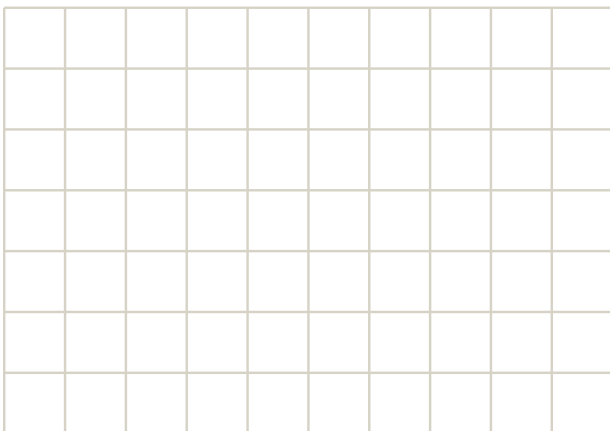
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Draw your own.

2 QUESTIONS

REMEMBER Large features may cover several squares.

1 Draw a map of your school on the grid. Mark 5 places and list their grid references.



2 Write directions from one place to another using grid references.

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

A treasure is 3 squares right and 4 squares up from B2. Where is it?

1. 3 squares right of B is E
2. 4 squares up from 2 is 6

Answer: E6

YOU DO

On your own: read the worked example, then try these.

- E1** From H7, go 5 left and 2 down. Where are you?

- E2** How many squares is it from A1 to J10 if you can only move up or right?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

What is at A8?

Level 2

Write the grid reference of the lake.

Level 3

From D9 go 4 right and 2 down. What is there?

Lesson summary

Missed the lesson? Start here.

Remember

1. A grid reference names a square.
2. Letter first (across), then number (up).
3. Large features may cover several squares.
4. Check by running your finger across, then up.

Level 1

What is at F2?

1. Column F, row 2

Answer: the lake

Level 2

Library

1. Column D, row 9

Answer: D9

Level 3

Car park

1. Covers 4 squares

Answer: B2, C2, B3, C3

Extension

3 right, 4 up from B2

1. B → E, 2 → 6

Answer: E6

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.02

Points in the first quadrant

Date _____

Learning intentions

We are learning to:

- plot and read points on a number plane in the first quadrant

Success criteria

I can:

- ☐ know the **x-coordinate** is across and the **y-coordinate** is up
- ☐ plot a point (x, y)
- ☐ read the coordinates of a point

Key terms Write the definition as your teacher explains it.

Coordinates

Origin

Axis

Notes

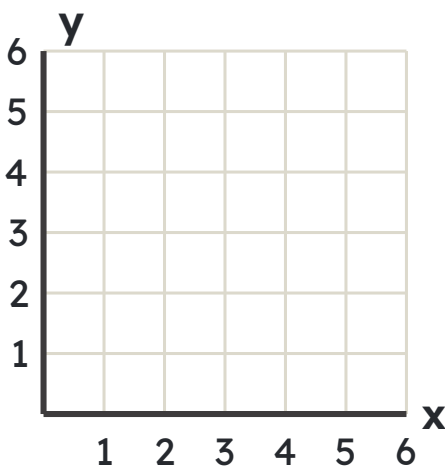
Level 1 · Getting started

Easy · 14 questions

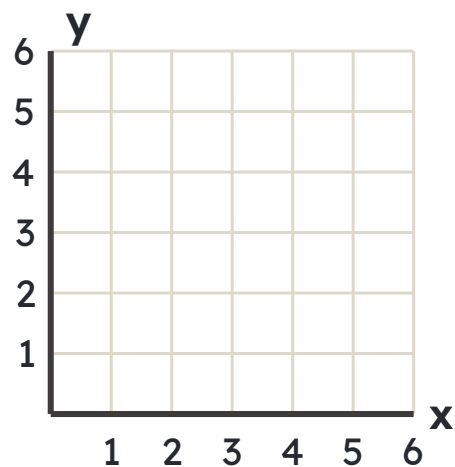
WE DO With your teacher: copy what they write on the board.

Example 1 Plot and label the points.

a Plot A(2, 3) and B(1, 6).



b Plot C(3, 0) and D(0, 4).

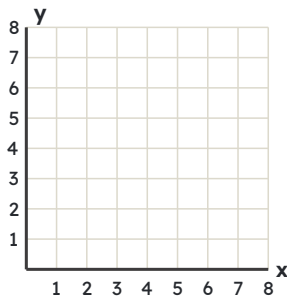


YOU DO On your own: **Easy** Set A. Plot and label each point on the same number plane.

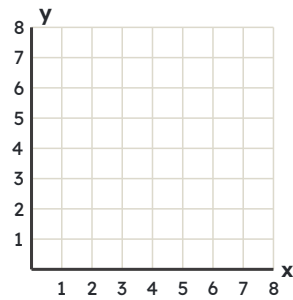
2 QUESTIONS

REMEMBER (x, y): x across first, then y up.

1 A(2, 3), B(1, 6), C(3, 0)



2 D(0, 4), E(5, 2), F(6, 5)



YOU DO On your own: **Easy** Set A2. Quick drill: write the x-coordinate (x) or the y-coordinate (y).

6 QUESTIONS

1 x of (4, 7)

2 y of (4, 7)

3 x of (0, 3)

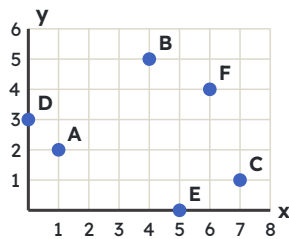
4 y of (6, 0)

5 y of (2, 9)

6 x of (8, 1)

YOU DO On your own: **Easy** Set B. Write the coordinates of each point.

6 QUESTIONS



1 A

2 B

3 C

4 D

5 E

6 F

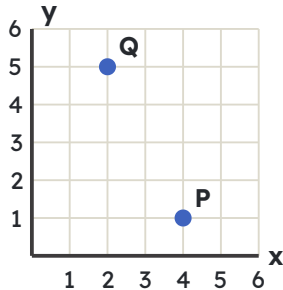
Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Read the coordinates, then explain.

a Write the coordinates of P and Q.



b Is (2, 5) the same point as (5, 2)? Explain.

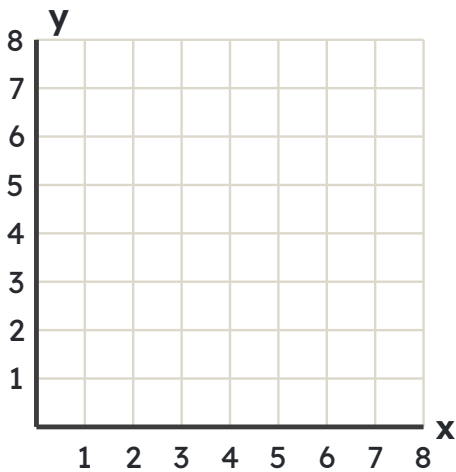
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Plot, join in order and name the shape.

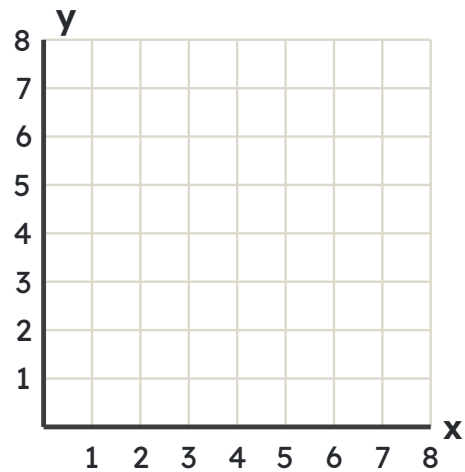
2 QUESTIONS

REMEMBER Start at the origin (0, 0).

1 (1, 1), (6, 1), (6, 4), (1, 4), (1, 1)



2 (2, 1), (6, 1), (4, 6), (2, 1)



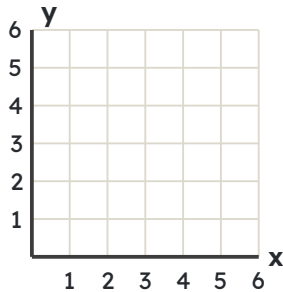
Level 3 · Challenge

Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Plot, join and name the shape.

- a** Plot $(1, 1)$, $(5, 1)$, $(5, 4)$, $(1, 4)$ and join them. Name the shape.



- b** What are the coordinates of the origin?

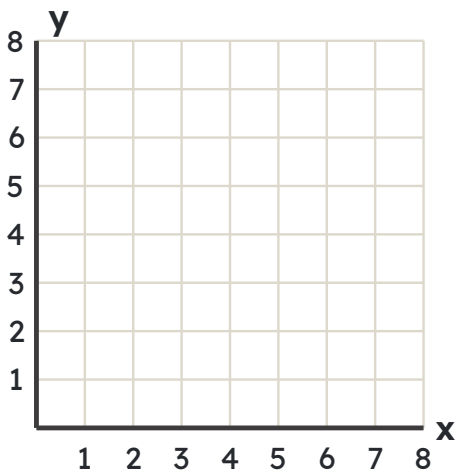
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

2 QUESTIONS

REMEMBER Points on the x-axis have $y = 0$.

- 1** Three corners of a square are $(1, 2)$, $(5, 2)$ and $(5, 6)$. Plot them and find the fourth corner.



- 2** Name 3 points on the x-axis and 3 on the y-axis. What do you notice?

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Find the midpoint of (2, 3) and (6, 7).

1. Halfway across: $(2 + 6) \div 2 = 4$
2. Halfway up: $(3 + 7) \div 2 = 5$

Answer: (4, 5)

YOU DO

On your own: read the worked example, then try these.

- E1 Find the midpoint of (1, 8) and (5, 2).

- E2 A rectangle has corners (1, 1) and (7, 5). Find its area in square units.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

What is the x-coordinate of (7, 3)?

Level 2

Where is the point (0, 5)?

Level 3

A rectangle has corners (1, 2), (6, 2), (6, 5). Find the fourth corner.

Lesson summary

Missed the lesson? Start here.

Remember

1. (x, y) : x across first, then y up.
2. Start at the origin $(0, 0)$.
3. Points on the x -axis have $y = 0$.
4. Points on the y -axis have $x = 0$.

Level 1

Plot $A(2, 3)$

1. 2 across, 3 up

Answer: a dot labelled A

Level 2

Coordinates of P

1. 4 across, 1 up

Answer: $(4, 1)$

Level 3

$(2, 5)$ vs $(5, 2)$

1. Order matters

Answer: different points

Extension

Midpoint of $(2, 3)$, $(6, 7)$

1. Average x and y

Answer: $(4, 5)$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.03

The number plane

Date _____

Learning intentions

We are learning to:

- plot and read points in all four quadrants

Success criteria

I can:

- ☐ name the four quadrants
- ☐ plot points with negative coordinates
- ☐ state which quadrant a point is in from its signs

Key terms Write the definition as your teacher explains it.

Quadrant

Origin

Notes

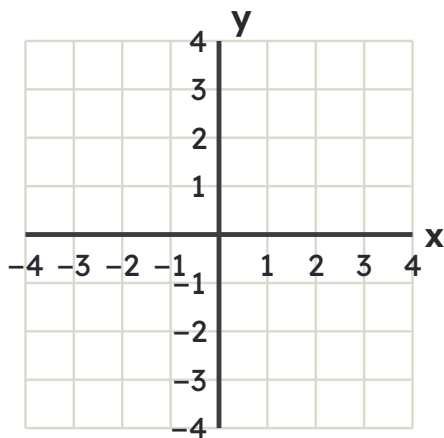
Level 1 · Getting started

Easy · 21 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Label the quadrants, then name the quadrant.

a Label the quadrants 1st to 4th.



b In which quadrant is $(-3, -3)$?

Your teacher works this out. Copy every step.

YOU DO

On your own: **Easy** Set A. Questions 1–6: Which quadrant or axis is each point on?

15 QUESTIONS

REMEMBER The axes cross at the origin (0, 0).

1 (–3, –3)

2 (4, 2)

3 (–1, 5)

4 (3, –6)

5 (0, 4)

6 (–2, 0)

Questions 7–15: which quadrant or axis?

7 (2, –5)

8 (–4, –1)

9 (–3, 6)

10 (0, –2)

11 (5, 5)

12 (–7, 0)

13 (1, –1)

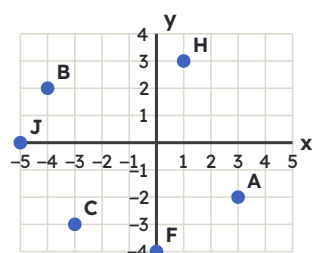
14 (–2, 3)

15 (0, 0)

YOU DO

On your own: **Easy** Set B. Write the coordinates of each point.

6 QUESTIONS



1 H

2 B

3 C

4 A

5 F

6 J

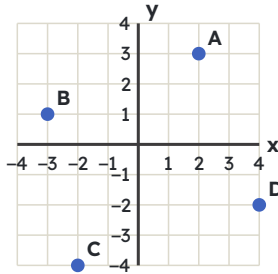
Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Read the coordinates, then name the quadrant.

a Write the coordinates of each point.



b Which quadrant has a positive x-coordinate and a negative y-coordinate?

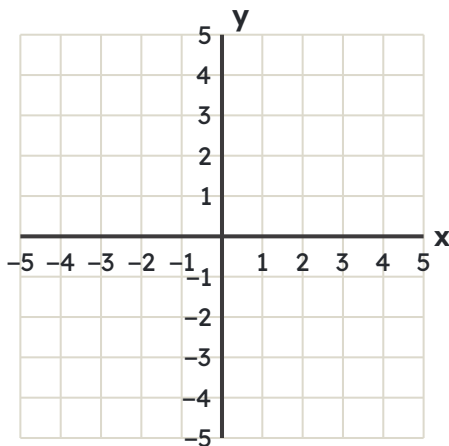
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Plot and join.

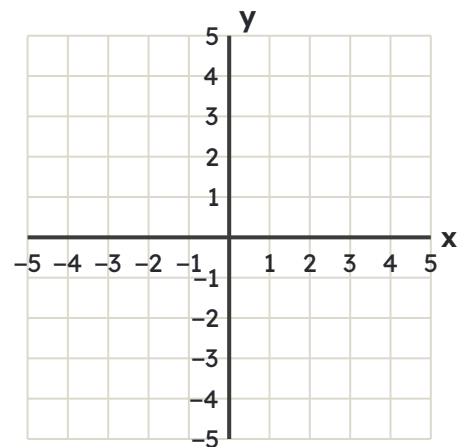
2 QUESTIONS

REMEMBER 1st (+, +), 2nd (-, +), 3rd (-, -), 4th (+, -).

1 (1, 3), (2, 3), (2, 2), (3, 1), (2, 0), (2, -1), (1, -1), (0, -2), (-1, -1), (-2, -1), (-2, 0), (-3, 1), (-2, 2), (-2, 3), (-1, 3), (0, 4), (1, 3). What shape is it?



2 Plot A(-4, -2), B(2, -2), C(2, 3). Find D so ABCD is a rectangle.



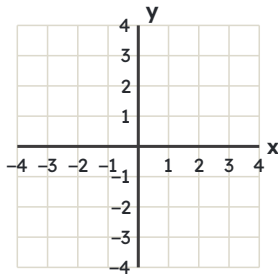
Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Plot and join, then write points.

- a** Plot $(1, 3)$, $(2, 2)$, $(3, 1)$, $(2, 0)$, $(1, -1)$, $(0, -2)$, $(-1, -1)$ and join.



- b** Write a point in the 2nd quadrant and a point on the y-axis.

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Think it through.

4 QUESTIONS

REMEMBER Quadrants go anticlockwise from top right.

- 1** In which quadrant is a point with a negative x-coordinate and a positive y-coordinate? A 1st B 2nd C 3rd D 4th

- 3** Reflect $(3, -2)$ in the y-axis. Write the new point.

- 2** What are the coordinates of the origin?

- 4** What is the distance between $(-3, 2)$ and $(4, 2)$?

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

A square has corners $(-2, -1)$ and $(3, -1)$ along its bottom. Find the other two corners (above).

1. Side length = $3 - (-2) = 5$
2. Go up 5: $y = -1 + 5 = 4$

Answer: $(3, 4)$ and $(-2, 4)$

YOU DO

On your own: read the worked example, then try these.

- E1 A triangle has corners $(-3, 0)$, $(3, 0)$ and $(0, 4)$. Find its area.

- E2 Reflect the triangle $(1, 1)$, $(4, 1)$, $(1, 3)$ in the x-axis. Write the new corners.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

In which quadrant is $(-4, 2)$?

Level 2

Write the coordinates of a point in the 4th quadrant.

Level 3

Reflect $(-2, 5)$ in the x-axis.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. The axes cross at the origin (0, 0).
 2. 1st (+, +), 2nd (–, +), 3rd (–, –), 4th (+, –).
 3. Quadrants go anticlockwise from top right.
 4. Points on an axis are in no quadrant.

Level 1

(–3, –3)

1. Both negative

Answer: 3rd quadrant

Level 2

Point B

1. 3 left, 1 up

Answer: (–3, 1)

Level 3

+x, –y

1. Bottom right

Answer: 4th quadrant

Extension

Square on (–2, –1), (3, –1)

1. Side 5, go up

Answer: (3, 4), (–2, 4)

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
9.04

Plotting tables of values

Date _____

Learning intentions

We are learning to:

- plot points from a table and describe the pattern

Success criteria

I can:

- ☐ read (x, y) pairs from a table
- ☐ plot the points and join them with a ruler
- ☐ describe the pattern (straight line, vertical, horizontal, increasing, decreasing)

Key terms Write the definition as your teacher explains it.

Table of values

Pattern

Notes

Level 1 · Getting started

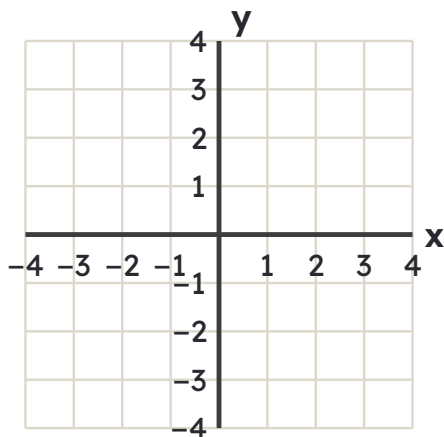
Easy · 12 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Plot the table, then describe the pattern.

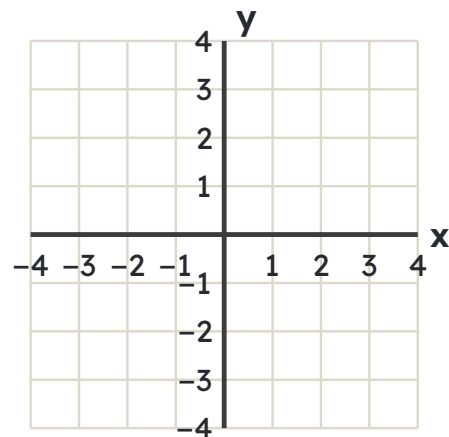
a Plot the table and describe the pattern.

x	-2	-2	-2	-2
y	-2	-1	0	1



b Plot and describe.

x	-2	-1	0	2
y	3	3	3	3



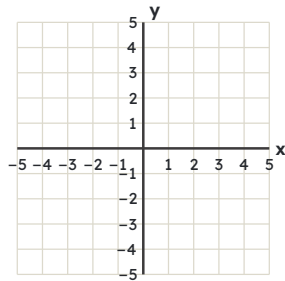
YOU DO On your own: **Easy** Set A. Plot each table and join the points.

2 QUESTIONS

REMEMBER Each column of a table is a point (x, y).

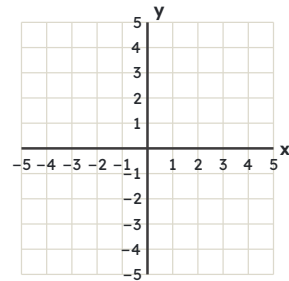
1

x	-2	-2	-2	-2
y	-2	-1	0	1



2

x	-2	-1	0	2
y	3	3	3	3



YOU DO On your own: **Easy** Set A2. Quick drill: write each table column as a point (x, y).

6 QUESTIONS

1 $x = 0, y = 3$

2 $x = 2, y = -1$

3 $x = -4, y = 0$

4 $x = -1, y = -5$

5 $x = 5, y = 5$

6 $x = 0, y = 0$

YOU DO On your own: **Easy** Set B. Write the table as a list of points.

4 QUESTIONS

1

x	0	1	2
y	5	4	3

3

x	1	2	3
y	2	4	6

2

x	-1	0	1
y	-1	1	3

4

x	0	2	4
y	1	1	1

Level 2 · Building up

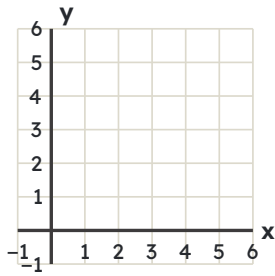
Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Plot the table, then describe the pattern.

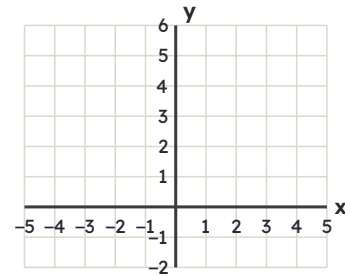
a Plot and describe.

x	0	1	2	5
y	5	4	3	0



b Plot and describe.

x	-1	0	1	2
y	-1	1	3	5



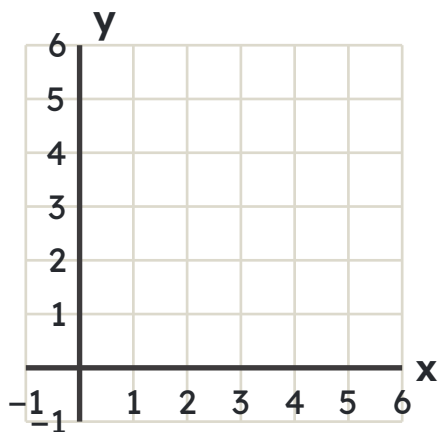
YOU DO On your own: **Medium** Set C. Plot, join and describe the pattern.

2 QUESTIONS

REMEMBER Plot each point carefully.

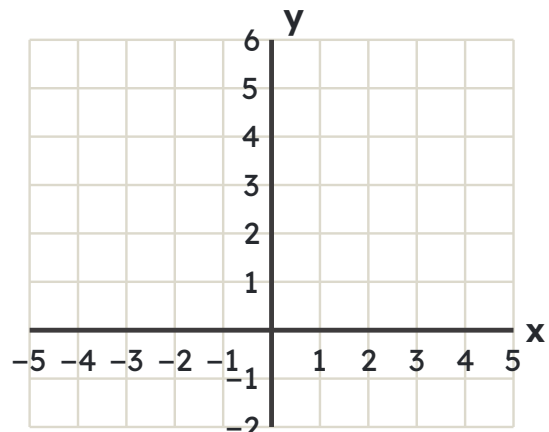
1

x	0	1	2	5
y	5	4	3	0



2

x	-1	0	1	2
y	-1	1	3	5



Level 3 · Challenge

Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Find the pattern, then predict.

a In the table in Example 4, what happens to y when x goes up by 1?

Your teacher works this out. Copy every step.

b Predict y when $x = 3$ in Example 4.

Your teacher works this out. Copy every step.

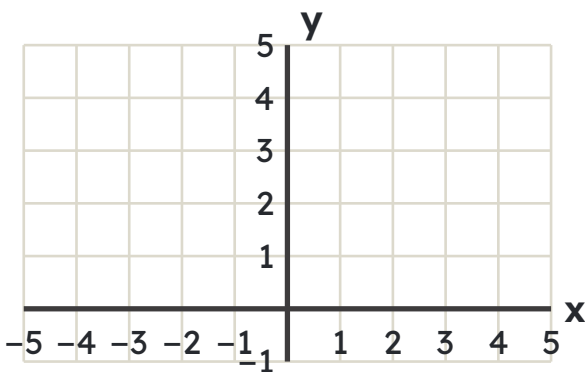
YOU DO On your own: **Challenging** Set D. Plot and describe.

2 QUESTIONS

REMEMBER Join them with a ruler if they line up.

1

x	-2	-1	0	1	2
y	4	1	0	1	4



2 Is the pattern in Question 1 a straight line? Describe it.

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

The points (0, 2), (1, 5), (2, 8) follow a pattern.
Find y when $x = 10$.

1. y goes up by 3 each time x goes up by 1
2. Start at 2 when $x = 0$: $y = 3x + 2$
3. $3 \times 10 + 2$

Answer: 32

YOU DO

On your own: read the worked example, then try these.

- E1 The points (0, 7), (1, 5), (2, 3) follow a pattern.
Find y when $x = 6$.

- E2 Find the rule for the points (1, 4), (2, 8), (3, 12).

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

Write the points in this table: $x = 0, 1, 2$ and $y = 3, 5, 7$.

Level 2

Describe the pattern of the points (1, 4), (2, 4), (3, 4).

Level 3

What happens to y each time x goes up by 1 for (0, 1), (1, 4), (2, 7)?

Lesson summary

Missed the lesson? Start here.

Remember

- Each column of a table is a point (x, y).
- Plot each point carefully.
- Join them with a ruler if they line up.
- Describe: horizontal, vertical, going up or going down.

Level 1

x always -2

- All points above/below each other

Answer: vertical line

Level 2

y always 3

- All points level

Answer: horizontal line

Level 3

(0, 5), (1, 4), (2, 3)

- y goes down 1

Answer: line going down

Extension

(0, 2), (1, 5), (2, 8)

- $y = 3x + 2$

Answer: 32 when $x = 10$

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
9.05

Tables of values from rules

Date _____

Learning intentions

We are learning to:

- complete a table of values from a rule

Success criteria

I can:

- ☐ substitute each x value into the rule
- ☐ use negative numbers carefully
- ☐ write the rule from a table

Key terms Write the definition as your teacher explains it.

Rule

Substitute

Notes

Level 1 · Getting started

Easy · 15 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Complete each table from its rule.

a Complete for $y = x + 3$.

x	-1	0	1	2	3
y					

b Complete for $q = 2p - 7$.

p	-2	-1	0	1	2
q					

Your teacher works this out. Copy every step.

Your teacher works this out. Copy every step.

YOU DO On your own: **Easy** Set A. Complete each table.

4 QUESTIONS

REMEMBER Write the rule.

① $y = x + 3$

x	-1	0	1	2	3
y					

③ $y = x - 4$

x	0	1	2	3	4
y					

② $y = 2x$

x	-1	0	1	2	3
y					

④ $y = 5x$

x	-2	-1	0	1	2
y					

YOU DO On your own: **Easy** Set A2. Quick drill: find y.

9 QUESTIONS

① $y = x + 4, x = 2$

② $y = 3x, x = 5$

③ $y = x - 7, x = 1$

④ $y = 2x + 1, x = 3$

⑤ $y = 10 - x, x = 4$

⑥ $y = 4x - 2, x = 0$

⑦ $y = -x, x = -3$

⑧ $y = 2x - 5, x = -1$

⑨ $y = x \div 2, x = 8$

YOU DO On your own: **Easy** Set B. Complete each table.

2 QUESTIONS

① $q = 2p - 7$

p	-2	-1	0	1	2
q					

② $d = 5 - c$

c	-1	0	1	2	3
d					

Level 2 · Building up

Medium · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Complete each table from its rule.

a Complete for $d = 5 - c$.

c	-1	0	1	2	3
d					

Your teacher works this out. Copy every step.

b Complete for $y = 3x + 1$.

x	-2	-1	0	1	2
y					

Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Complete each table.

4 QUESTIONS

REMEMBER Replace x with each value (use brackets for negatives).

1 $y = 3x + 1$

x	-2	-1	0	1	2
y					

3 $y = \frac{1}{2}x - 2$

x	-4	-2	0	2	4
y					

2 $y = -2x + 3$

x	-2	-1	0	1	2
y					

4 $y = 10 - 3x$

x	0	1	2	3	4
y					

Level 3 · Challenge

Challenging · 4 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Find the rule for each table.

a Find the rule.

x	0	1	2	3
y	4	6	8	10

Your teacher works this out. Copy every step.

b Find the rule.

x	1	2	3	4
y	7	6	5	4

Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Find the rule for each table.

4 QUESTIONS

REMEMBER Calculate y.

1

x	0	1	2	3
y	4	6	8	10

3

x	0	1	2	3
y	-3	1	5	9

2

x	1	2	3	4
y	7	6	5	4

4

x	-1	0	1	2
y	3	0	-3	-6

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Complete a table for $y = x^2 - 1$ using $x = -2$ to 2.

1. Square first, then subtract 1
2. $(-2)^2 - 1 = 3$

Answer: $y = 3, 0, -1, 0, 3$

YOU DO

On your own: read the worked example, then try these.

- E1 Complete a table for $y = x^2 + x$ using $x = -2$ to 2.

- E2 Find the rule: $x = 1, 2, 3, 4$ and $y = 1, 4, 9, 16$.

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

For $y = x + 5$, find y when $x = -2$.

Level 2

For $y = 4x - 1$, complete for $x = 0, 1, 2$.

Level 3

Find the rule: $x = 0, 1, 2$ and $y = 1, 4, 7$.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Write the rule.
 2. Replace x with each value (use brackets for negatives).
 3. Calculate y .
 4. To find a rule: how much does y change when x goes up 1? Then find the start value.

Level 1

$$y = x + 3$$

1. $-1 + 3 = 2, \dots$

Answer: 2, 3, 4, 5, 6

Level 2

$$q = 2p - 7$$

1. $2 \times (-2) - 7 = -11$

Answer: -11, -9, -7, -5, -3

Level 3

Rule for 4, 6, 8, 10

1. $+2$ each time, start 4

Answer: $y = 2x + 4$

Extension

$$y = x^2 - 1$$

1. Square first

Answer: 3, 0, -1, 0, 3

My reflection: what did I learn today? Which question was hardest, and why?

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LESSON
9.06

Graphing linear equations

Date _____

Learning intentions

We are learning to:

- graph a linear equation using a table of values
- test whether a point lies on a line

Success criteria

I can:

- ☐ complete a table of values for the equation
- ☐ plot the points and rule a straight line through them
- ☐ substitute a point into the equation to test it

Key terms Write the definition as your teacher explains it.

Linear

Equation

Notes

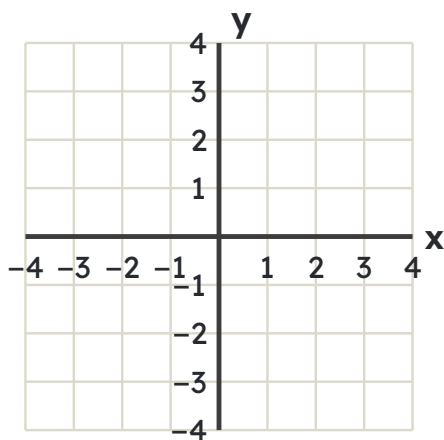
Level 1 · Getting started

Easy · 14 questions

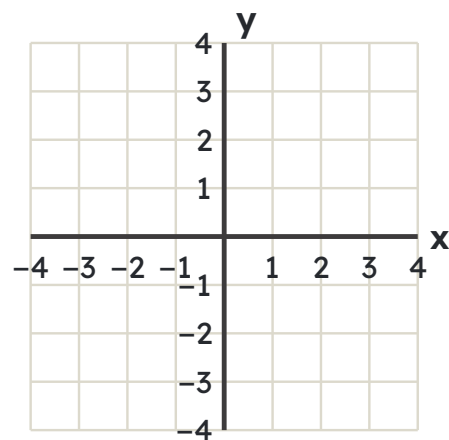
WE DO With your teacher: copy what they write on the board.

Example 1 Complete a table, then graph the line.

a Graph $y = 2x - 1$ for $x = -1$ to 2.



b Graph $y = -x + 3$.

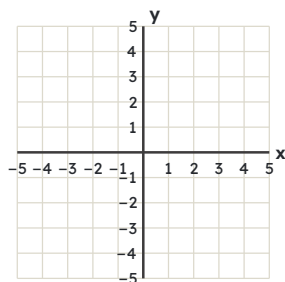


YOU DO On your own: **Easy** Set A. Complete a table, then graph.

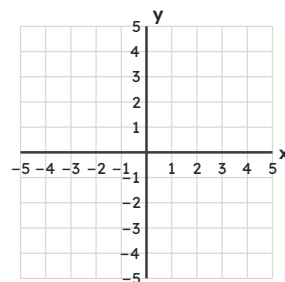
2 QUESTIONS

REMEMBER Make a table with 3 or more x values.

1 $y = x + 1$ ($x = -2$ to 2)



2 $y = 2x - 1$ ($x = -1$ to 2)



YOU DO On your own: **Easy** Set A2. Quick drill: does the point lie on $y = 2x + 1$? Write yes or no.

6 QUESTIONS

1 $(0, 1)$

2 $(1, 3)$

3 $(2, 4)$

4 $(-1, -1)$

5 $(3, 7)$

6 $(-2, -4)$

YOU DO On your own: **Easy** Set B. Does the point lie on $y = 3x + 2$? Write yes or no.

6 QUESTIONS

1 $(1, 2)$

4 $(0, 2)$

2 $(-1, 4)$

5 $(1, 5)$

3 $(-2, -4)$

6 $(2, 8)$

Level 2 · Building up

Medium · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 2 Does the point lie on the line?

a Does $(1, 5)$ lie on $y = 3x + 2$?

Your teacher works this out. Copy every step.

b Does $(-2, -4)$ lie on $y = 3x + 2$?

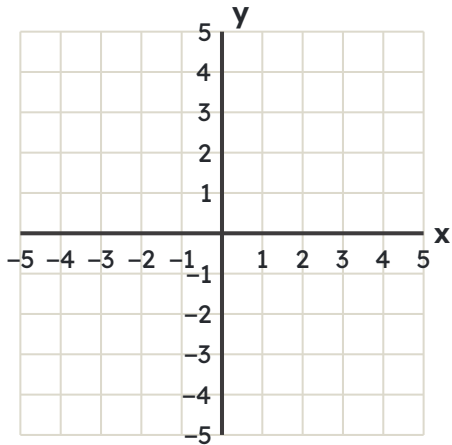
Your teacher works this out. Copy every step.

YOU DO On your own: **Medium** Set C. Complete a table, then graph.

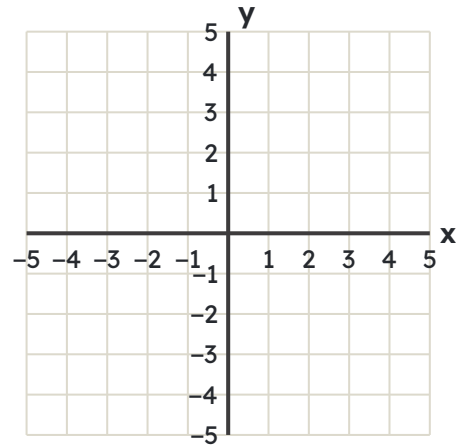
2 QUESTIONS

REMEMBER Substitute to find y .

1 $y = -x + 3$



2 $y = \frac{1}{2}x - 2$



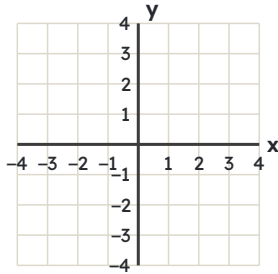
Level 3 · Challenge

Challenging · 2 questions

WE DO With your teacher: copy what they write on the board.

Example 3 Graph the line, then answer.

a Graph $y = \frac{1}{2}x - 2$ for $x = -4$ to 4 .



b Where does $y = 2x - 1$ cross the y-axis?

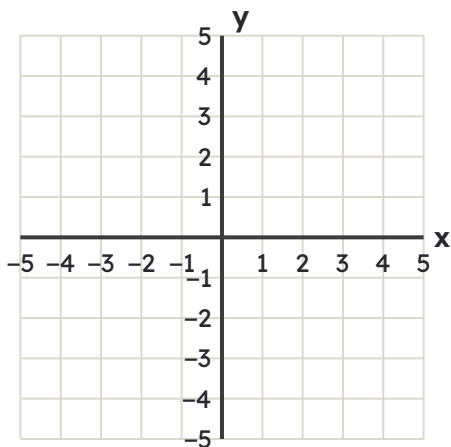
Your teacher works this out. Copy every step.

YOU DO On your own: **Challenging** Set D. Graph and compare.

2 QUESTIONS

REMEMBER Plot and rule a line through all the points.

1 Graph $y = x$ and $y = x + 2$ on the same plane. What do you notice?



2 Where do $y = x + 1$ and $y = -x + 3$ meet? (Use your graphs.)

Extension · Stretch yourself

Finished Set D? Stretch yourself here.

Worked example

Find the value of k if $(2, k)$ lies on $y = 4x - 3$.

1. Substitute $x = 2$
2. $k = 4 \times 2 - 3$

Answer: $k = 5$

YOU DO

On your own: read the worked example, then try these.

- E1 Find m if $(m, 11)$ lies on $y = 2x + 5$.

- E2 Which of these lines are parallel: $y = 2x + 1$, $y = x + 2$, $y = 2x - 3$?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose ONE question. Show your working.

Name _____ Class _____

Level 1

For $y = x + 4$, find y when $x = 1$.

Level 2

Does $(2, 5)$ lie on $y = 2x + 1$?

Level 3

Does $(-1, 3)$ lie on $y = -2x + 1$?

Lesson summary

Missed the lesson? Start here.

Remember

1. Make a table with 3 or more x values.
2. Substitute to find y .
3. Plot and rule a line through all the points.
4. Test a point: substitute x and see if you get its y .

Level 1

$$y = 2x - 1$$

1. $x: -1, 0, 1, 2 \rightarrow y: -3, -1, 1, 3$

Answer: a straight line through $(0, -1)$

Level 2

$$(1, 5) \text{ on } y = 3x + 2?$$

1. $3 \times 1 + 2 = 5 \checkmark$

Answer: yes

Level 3

$$(-2, -4)$$

1. $3 \times (-2) + 2 = -4 \checkmark$

Answer: yes

Extension

$$(2, k) \text{ on } y = 4x - 3$$

1. $8 - 3$

Answer: $k = 5$

My reflection: what did I learn today? Which question was hardest, and why?

This space is left blank on purpose. It is the back of your exit ticket.